

Coding & Solution

Date	01 July 2026
Team ID	[Enter Team ID]
Project Name	HDI Predictor
Maximum Marks	5 Marks

Coding & Solution

This section evaluates the implementation quality of the HDI Predictor project. The project is developed using Machine Learning and Flask to automate the country human development index calculation and prediction process. The code is modular, well-documented, follows Python coding standards, and satisfies the functional requirements of the system.

Solution Summary

Field	Details
Repository Link / URL	https://github.com/charanemandi26-arch/Human-Development-Index-Prediction
Programming Language(s)	Python 3.11, HTML5, CSS3
Framework(s) Used	Flask, Scikit-learn, Pandas, NumPy, Pickle
Key Features Implemented	• Data preprocessing and cleaning • Feature scaling and scaling verification • Model training using Linear Regression, Random Forest, and Gradient Boosting • Best model saved using Pickle • Flask-based web application for real-time prediction • User-friendly input form for country indicators • Instant HDI score and development tier prediction
Pending / Incomplete Features	• User login authentication • Database integration for storing prediction history • Email/SMS notification system • Cloud deployment (optional enhancement)
Setup / Run Instructions	1. Install Python libraries using <code>pip install -r requirements.txt</code>. 2. Place <code>scaler.pkl</code> and <code>hdi_model.pkl</code> in the <code>model/</code> folder. 3. Run <code>python app.py</code>. 4. Open <code>http://127.0.0.1:5000</code> in a web browser. 5. Enter country indicator details and click Predict to view the result.

Code Quality Checklist

S.No	Criteria	Status(yes/no)
1	Code is modular and organized into functions / classes	Yes
2	Meaningful variable and function names are used	Yes
3	Code includes comments / documentation where necessary	Yes
4	Error handling is implemented for critical operations	Yes
5	The application runs without critical errors	Yes
6	Code is committed to a version control repository	Yes

Additional Notes / Comments

The HDI Predictor project follows good software engineering practices by maintaining a modular architecture, reusable code components, and clear documentation. Machine learning algorithms are trained and evaluated to select the best-performing model, which is integrated into a Flask web application for real-time prediction. The application is scalable and can be enhanced further by adding database support, authentication, cloud deployment, and prediction history management.